DECODELABS DATA ANALYTICS INTERNSHIP

PROJECT 1: DATA CLEANING & PREPARATION

1. Project Overview

Project: Data Cleaning & Preparation

Organisation: Decodelabs

Tool Used: Microsoft Excel 2021

Dataset: E-commerce Dataset

Original Dataset Size: 1,200 records and 14 columns.

Project Objective:

To inspect, clean and prepare the raw dataset for analysis by identifying and handling missing values, checking for duplicates, correcting data formats and validating the overall quality of the dataset.

2. Dataset Preparation

The original dataset was converted into an Excel Workbook (.xlsx) format.

The following worksheets were created:

1. Raw Data: Preserved the original dataset without making changes.

2. Cleaned_Data: Used as the working sheet for cleaning and preparing the dataset.

3. Data Quality Check Sheet: Used to perform and record validation checks.

4. Cleaning Log: Created to document identified issues, affected records and cleaning actions.

The dataset contained 14 columns:

- Order ID
- Date
- Customer ID
- Product
- Quantity
- Unit Price
- Shipping Address
- Payment Method
- Order Status
- Tracking Number
- Items in Cart
- Coupon Code
- Referral Source

- Total Price

3. Data Inspection and Cleaning Activities

The dataset was inspected using Excel filters and other Excel functions to identify missing values, duplicates, inconsistent entries and formatting issues.

The following activities were performed:

A. Handling Missing Values

- Identified 309 blank cells in the Coupon Code column.
- Used an IF formula to identify blank coupon entries and replace them with "No Coupon".
- Applied Paste Special → Values to retain the results without leaving formulas in the cleaned data.
- Checked the other columns for missing values.

B. Duplicate Checks

- Inspected the dataset for duplicate records.
- Performed a separate validation of Order IDs to identify repeated IDs.
- No duplicate Order IDs were detected.

C. Date Formatting

- Applied a custom date format to the Date column.
- Used the format DD MMM-YYYY to display dates consistently, for example, 04 Jan-2023.
- Checked the date entries for numeric Excel values.

D. Currency Formatting

- Applied Currency formatting to the Unit Price column.
- Applied Currency formatting to the Total Price column.

E. Data Consistency and Value Inspection

- Used Excel filters to inspect the dataset for errors and inconsistencies.
- The following fields were checked:

Customer ID

Product

Quantity

Shipping Address

Payment Method

Order Status

Tracking Number

Items in Cart

Coupon Code

Referral Source

Other dataset fields.

No obvious inconsistencies were identified in the reviewed fields.

F. Numerical Value Checks

- Checked Quantity for negative, zero and unreasonable values.
- Checked Unit Price and Total Price for negative, zero and unreasonable values.
- No such values were identified during the inspection.

4. Data Quality Assessment and Validation

After the cleaning activities, a separate Data Quality Check Sheet was used to validate the cleaned dataset.

Three major validation checks were performed using Excel formulas.

Check 1: Duplicate Order IDs

Used a SUMPRODUCT and COUNTIF formula to identify duplicate Order IDs in the Cleaned_Data worksheet.

Result: 0

Interpretation: No duplicate Order IDs were detected.

Check 2: Valid Date Count

Used the COUNT function to check the number of numeric date entries in the Date column.

Formula:

=COUNT(Cleaned_Data!B2:B1201)

Result: 1,200

Interpretation: All 1,200 date cells contain numeric values recognised by Excel.

Check 3: Remaining Blank Cells

Used the COUNTBLANK function to check for empty cells across the 14 dataset columns.

Formula: =COUNTBLANK(Cleaned_Data!A2:N1201)

Result: 0

Interpretation: No blank cells remained in the cleaned dataset.

5. Data Quality Check Results

Validation Check

Duplicate Order IDs

Numeric Date Entries

Remaining Blank Cells

All three recorded validation checks returned their expected results.

6. Documentation

A Cleaning Log was created to maintain a record of:

- Data quality issues identified.
- Columns affected.
- Number of rows affected, where applicable.
- Cleaning actions performed.

Final validation results.

A separate Data Quality Check Sheet was also maintained to provide measurable evidence of the cleaning results.

7. Project Outcome

The dataset was inspected, cleaned and subjected to data quality checks using Microsoft Excel 2021.

The cleaning process addressed the 309 missing Coupon Code entries, standardised the display of date and currency fields, and included checks for duplicate Order IDs, missing values and unreasonable numerical entries.

The final recorded validation results were:

Zero duplicate Order IDs.

1,200 numeric date entries.

Zero remaining blank cells.

Conclusion:

The Excel data cleaning and preparation stage has been completed based on the activities and validation checks documented above. The cleaned dataset is prepared for the next stage of the data analytics project, subject to any additional date-range or date-validity checks required by the project brief.